Chapter 3. Inference for Stationary ARMA Models

Estimation of mean and ACF, Yule-Walker, LS, ML, prediction

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3.1 Estimation of the mean

Motivation. For i.i.d. data the central limit theorem gives $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, the basis of tests and confidence intervals. For a stationary ARMA series the same holds, but the asymptotic variance is a function of the model parameters and can differ greatly from $\mathrm{Var}(X_t)$.

Throughout, the subscript n on statistics is dropped ($\bar{X} = \bar{X}_n$, $\hat{\theta} = \hat{\theta}_n$), and limits are as $n \to \infty$. Consider the AR(1) model

$$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad |\phi| < 1.$$

By (2.5),

$$X_t = \mu + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}. \tag{3.1}$$

Theorem 3.1.1 (CLT for the sample mean of an AR(1))

If $\{X_t\}$ is the stationary series (3.1) and $\hat{\mu} = \bar{X} = n^{-1}\sum_{t=1}^{n} X_t$, then

$$\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N\Big(0, \frac{\sigma^2}{(1-\phi)^2}\Big).$$

Proof.

Step 1. (Decomposition.) Put $b_j = \sum_{k \geq j+1} \phi^k = \phi^{j+1}/(1-\phi)$ and $y_t = \sum_{j \geq 0} b_j \epsilon_{t-j}$. Since $\sum |b_j| < \infty$, $\{y_t\}$ is stationary (Lemma 2.2.A). As $b_{j-1} - b_j = \phi^j$ and $b_0 = \phi/(1-\phi)$,

$$\begin{aligned} X_t - \mu &= \epsilon_t + \sum_{j \geq 1} (b_{j-1} - b_j)\epsilon_{t-j} = \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - b_0\epsilon_t + \sum_{j \geq 0} b_j \epsilon_{t-1-j} - \sum_{j \geq 1} b_j \epsilon_{t-j} \\ &= \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - y_t + y_{t-1}. \end{aligned}$$

Step 2. (Telescoping.) Summing over t,

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} (X_t - \mu) = \underbrace{\frac{1}{1-\phi} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \epsilon_t}_{\text{(I)}} \underbrace{- \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0}_{\text{(II)}}.$$

Step 3. By the i.i.d. CLT, (I) $\xrightarrow{d} N(0, \sigma^2/(1-\phi)^2)$, since $\sum_j \phi^j = 1/(1-\phi)$.

Step 4. $E(\text{II})^2 \leq \frac{2}{n}(Ey_n^2 + Ey_0^2) = \frac{4}{n} Ey_0^2 \to 0$, so (II) $\xrightarrow{p} 0$ by Chebyshev's inequality. Slutsky's theorem gives the claim. ∎

Remark 3.1.2 (General linear processes, ARMA)

The same argument applies to any $X_t = \mu + \sum_{j \geq 0} a_j \epsilon_{t-j}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, with $|a_j| \leq c\rho^j$ ($c > 0$, $0 < \rho < 1$); by Proposition 2.4.A this covers every causal ARMA(p, q). Put $b_j = \sum_{k \geq j+1} a_k$; then $|b_j| \leq c\rho^{j+1}/(1-\rho)$, so $\sum_j |b_j| < \infty$ and $y_t = \sum_j b_j \epsilon_{t-j}$ is stationary. Exactly as above, $X_t - \mu = (\sum_j a_j)\epsilon_t - y_t + y_{t-1}$, hence

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}}\Big(\sum_{j=0}^{\infty} a_j\Big) \sum_{t=1}^{n} \epsilon_t - \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0 \xrightarrow{d} N\Big(0, \Big(\sum_{j=0}^{\infty} a_j\Big)^2 \sigma^2\Big).$$

So the CLT survives without independence, but with a different variance. For testing $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ at level α: under i.i.d. data reject when $|\sqrt{n}\bar{X}| \geq z_{\alpha/2}\sigma$; for a linear process reject when $|\sqrt{n}\bar{X}| \geq z_{\alpha/2}\,|\sum_j a_j|\,\sigma$. Here $z_{\alpha/2}$ satisfies $P(|Z| \leq z_{\alpha/2}) = 1-\alpha$ for $Z \sim N(0,1)$. For AR(1), $\sum_j a_j = 1/(1-\phi)$, which is huge when ϕ is near 1, so getting the variance right is essential. In particular, the $100(1-\alpha)\%$ confidence interval for μ in an AR(1) model is

$$\bar{X} - \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)} \leq \mu \leq \bar{X} + \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)}.$$

3.2 Sample autocovariance, autocorrelation and partial autocorrelation

Motivation. The ACF and PACF identify ARMA orders, but in practice they must be estimated. This section defines the sample versions and gives Bartlett's theorem, the asymptotic distribution of the sample ACF, which tells us how far a sample autocorrelation can stray from zero by chance.

For an i.i.d. sample $(X_1, Y_1), \ldots, (X_n, Y_n)$ the covariance is estimated by $\widehat{\mathrm{Cov}}(X,Y) = n^{-1}\sum_{j=1}^{n}(X_j - \bar{X})(Y_j - \bar{Y})$. Accordingly:

Definition 3.2.1 (Sample autocovariance function)

For $|h| < n$, the *sample autocovariance function* is

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(X_t - \bar{X})(X_{t+|h|} - \bar{X}).$$

For fixed h, $\hat{\gamma}(h)$ is the sample autocovariance coefficient. For stationary ARMA series $P(\hat{\gamma}(h) \to \gamma(h)) = 1$ (by the ergodic theorem), so $\hat{\gamma}(h)$ is consistent. For large h few pairs enter $\hat{\gamma}(h)$, so it is reliable only for small lags; a common rule of thumb is $|h| \leq \sqrt{n}$. The divisor n (rather than $n - |h|$) makes the matrix $[\hat{\gamma}(i-j)]$ nonnegative definite.

Definition 3.2.2 (Sample autocorrelation function)

The *sample autocorrelation function* (SACF), the estimator of $\rho(h)$, is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Like ρ, it satisfies $|\hat{\rho}(h)| \leq 1$.

Definition 3.2.3 (Sample partial autocorrelation)

For fixed h, the last component $\hat{\phi}_{hh}$ of the solution of

$$\begin{pmatrix} \hat{\rho}(0) & \hat{\rho}(1) & \cdots & \hat{\rho}(h-1) \\ \hat{\rho}(1) & \hat{\rho}(0) & \cdots & \hat{\rho}(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}(h-1) & \hat{\rho}(h-2) & \cdots & \hat{\rho}(0) \end{pmatrix} \begin{pmatrix} \hat{\phi}_{h1} \\ \hat{\phi}_{h2} \\ \vdots \\ \hat{\phi}_{hh} \end{pmatrix} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \\ \vdots \\ \hat{\rho}(h) \end{pmatrix}$$

is the *sample partial autocorrelation coefficient*; as a function of h it is the *sample partial autocorrelation function* (SPACF).

Theorem 3.2.4 (Bartlett)

Let $\{X_t\}$ be stationary with

$$X_t = \mu + \sum_{j=0}^{\infty} a_j \epsilon_{t-j}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad \sum_{j=0}^{\infty} |a_j| < \infty, \quad E\epsilon_t^4 < \infty.$$

Then for each fixed $h \geq 1$,

$$\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W), \qquad \hat{\boldsymbol{\rho}}(h) = (\hat{\rho}(1), \ldots, \hat{\rho}(h))^T, \quad \boldsymbol{\rho}(h) = (\rho(1), \ldots, \rho(h))^T,$$

where $W = (w_{ij})_{i,j=1}^{h}$ is given by *Bartlett's formula*

$$\begin{aligned} w_{ij} = \sum_{k=-\infty}^{\infty} \Big\{ & \rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j) + 2\rho(i)\rho(j)\rho^2(k) \\ & - 2\rho(i)\rho(k)\rho(k+j) - 2\rho(j)\rho(k)\rho(k+i) \Big\} \\ = \sum_{k=1}^{\infty} & \{\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)\}\{\rho(k+j) + \rho(k-j) - 2\rho(j)\rho(k)\}. \end{aligned}$$

Proof. [Supplement] We give the argument at the level of the stated assumptions; two standard limit theorems are quoted. Write $\eta = E\epsilon_t^4/\sigma^4$ and, without loss of generality, $\mu = 0$. Let $\gamma^*(h) = n^{-1}\sum_{t=1}^{n} X_t X_{t+h}$.

Step 1. (Fourth moments.) Since $E\epsilon_i\epsilon_j\epsilon_k\epsilon_l = \sigma^4(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + (\eta - 3)\sigma^4\delta_{ijkl}$, expanding the four series ($a_j = 0$ for $j < 0$) gives

$$\begin{aligned} E[X_t X_{t+p} X_s X_{s+q}] = {} & \gamma(p)\gamma(q) + \gamma(s-t)\gamma(s-t+q-p) + \gamma(s-t+q)\gamma(s-t-p) \\ & + (\eta - 3)\sigma^4 \sum_i a_i a_{i+p} a_{i+s-t} a_{i+s-t+q}. \end{aligned}$$

Step 2. (Covariances.) Summing over $s, t = 1, \ldots, n$ and grouping by $k = s - t$,

$$\begin{aligned} n\,\mathrm{Cov}(\gamma^*(p), \gamma^*(q)) = \sum_{|k|<n} \Big(1 - \frac{|k|}{n}\Big) \Big[& \gamma(k)\gamma(k+q-p) + \gamma(k+q)\gamma(k-p) \\ & + (\eta - 3)\sigma^4 \sum_i a_i a_{i+p} a_{i+k} a_{i+k+q} \Big]. \end{aligned}$$

All terms are absolutely summable in k (as $\sum|\gamma(k)| \leq \sigma^2(\sum|a_j|)^2$), so by dominated convergence the limit is

$$V_{pq} = (\eta - 3)\gamma(p)\gamma(q) + \sum_{k=-\infty}^{\infty} [\gamma(k)\gamma(k+q-p) + \gamma(k+q)\gamma(k-p)],$$

using $\sum_k \sum_i a_i a_{i+p} a_{i+k} a_{i+k+q} = \sigma^{-4}\gamma(p)\gamma(q)$.

Step 3. (Joint CLT for γ^*.) For the truncated process $X_t^{(m)} = \sum_{j=0}^{m} a_j \epsilon_{t-j}$, the vectors $(X_t^{(m)} X_{t+l}^{(m)})_{l=0}^{h}$ form a strictly stationary $(m+h)$-dependent sequence, so the CLT for m-dependent sequences (Hoeffding–Robbins) together with the Cramér–Wold device gives $\sqrt{n}(\boldsymbol{\gamma}_m^* - \boldsymbol{\gamma}_m) \xrightarrow{d} N(0, V^{(m)})$, where $V^{(m)} \to V$ as $m \to \infty$ by Step 2 applied to $a^{(m)}$. Applying the computation of Step 2 to the difference $X_t - X_t^{(m)}$ gives $\limsup_n n\,\mathrm{Var}(\gamma^*(l) - \gamma_m^*(l)) \leq C\sum_{j>m}|a_j| \to 0$ as $m \to \infty$. By the standard approximation lemma (Billingsley, Thm. 3.2), $\sqrt{n}(\boldsymbol{\gamma}^* - \boldsymbol{\gamma}) \xrightarrow{d} N(0, V)$, with $\boldsymbol{\gamma}^* = (\gamma^*(0), \ldots, \gamma^*(h))^T$.

Step 4. (From γ^* to $\hat{\gamma}$.) Expanding the product in Definition 3.2.1,

$$\hat{\gamma}(l) - \gamma^*(l) = -\frac{1}{n}\sum_{t=n-l+1}^{n} X_t X_{t+l} - \bar{X}\cdot\frac{1}{n}\sum_{t=1}^{n-l}(X_t + X_{t+l}) + \Big(1 - \frac{l}{n}\Big)\bar{X}^2.$$

The first term has L^1 norm $\le l\gamma(0)/n$. The other two are bounded by $C\bar{X}^2 + O_p(n^{-1})|\bar{X}|$, and $\bar{X} = O_p(n^{-1/2})$ by Remark 3.1.2. Hence $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*) \xrightarrow{p} 0$ and $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \xrightarrow{d} N(0, V)$.

Step 5. (Delta method.) $\hat{\rho}(i) = g_i(\hat{\boldsymbol{\gamma}})$ with $g_i(x) = x_i/x_0$, whose gradient at $\boldsymbol{\gamma}$ has entries $-\rho(i)/\gamma(0)$ (in x_0) and $1/\gamma(0)$ (in x_i). Hence $\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W)$ with

$$w_{ij} = \frac{V_{ij} - \rho(i)V_{0j} - \rho(j)V_{i0} + \rho(i)\rho(j)V_{00}}{\gamma(0)^2}.$$

The $(\eta - 3)$ terms contribute $(\eta-3)[\rho(i)\rho(j) - \rho(i)\rho(j) - \rho(j)\rho(i) + \rho(i)\rho(j)] = 0$, so W does not depend on η. With $T_{pq} = V_{pq}/\gamma(0)^2$ (without the η term) and the shifts $k \mapsto k + i$, one gets $T_{ij} = \sum_k[\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j)]$, $T_{0j} = 2\sum_k \rho(k)\rho(k+j)$, $T_{i0} = 2\sum_k \rho(k)\rho(k+i)$ and $T_{00} = 2\sum_k \rho^2(k)$, which yields the first form of w_{ij}.

Step 6. (Second form.) Expand the product in the second form, sum over $k \ge 1$, and use $\rho(-k) = \rho(k)$ to rewrite each one-sided sum as half of a two-sided sum (the $k = 0$ terms cancel); this reproduces the first form. ■

Corollary 3.2.A (White-noise and MA(q) bands; supplement)

(1) If $\{X_t\} \sim IID(0, \sigma^2)$ with $EX_t^4 < \infty$, then $\sqrt{n}\,\hat{\boldsymbol{\rho}}(h) \xrightarrow{d} N(0, I_h)$.
(2) If $\{X_t\}$ is MA(q), then for $i > q$, $\sqrt{n}\,\hat{\rho}(i) \xrightarrow{d} N\big(0,\, 1 + 2\sum_{k=1}^{q}\rho^2(k)\big)$.

Proof. [Supplement]

Step 1. For i.i.d. data $\rho(k) = 0$ for $k \ne 0$. In the second form of Bartlett's formula, the factor $\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)$ with $k \ge 1$, $i \ge 1$ is nonzero only for $k = i$, where it equals 1. Thus $w_{ij} = \mathbb{1}\{i = j\}$.

Step 2. If $i > q$ then $\rho(i) = 0$ and $\rho(k+i) = 0$ for $k \ge 1$, so $w_{ii} = \sum_{k\ge1}\rho^2(k-i) = \sum_{m=1-i}^{\infty}\rho^2(m) = \rho^2(0) + 2\sum_{k=1}^{q}\rho^2(k)$, since $\rho(m) = 0$ for $|m| > q$. ■

Hence, for white noise, about 95% of the $\hat{\rho}(h)$ lie within $\pm 1.96/\sqrt{n}$; sample autocorrelations outside these bands indicate genuine dependence, and for an MA(q) the bands at lags $> q$ widen by the factor $(1 + 2\sum_{k\le q}\rho^2(k))^{1/2}$.

3.3 The method of moments

Motivation. The simplest estimator equates population and sample autocorrelations. For AR models this leads to a linear system (the Yule–Walker equations); it also supplies starting values for maximum likelihood.

For MA and ARMA models the moment equations are nonlinear and awkward to solve, so we treat AR models only. Consider the stationary (causal) AR(p) model

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2).$$

By causality X_{t-j} is uncorrelated with ϵ_t for $j \ge 1$, so

$$E\big[(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p})X_{t-j}\big] = E\epsilon_t X_{t-j} = 0, \qquad j = 1, \ldots, p,$$

which gives the *Yule–Walker equations* $\gamma(j) = \phi_1\gamma(j-1) + \cdots + \phi_p\gamma(j-p)$, $j = 1, \ldots, p$; in matrix form

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(p-1) \\ \rho(1) & \rho(0) & \cdots & \rho(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p) \end{pmatrix}. \tag{3.2}$$

Replacing every $\rho(j)$ by $\hat{\rho}(j)$ and solving gives the moment estimator $\hat{\boldsymbol{\phi}}_{YW}$ of $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)^T$, the *Yule–Walker (Y-W) estimator*. Comparing with Definition 3.2.3, the last component of the order-h Y-W estimator is exactly $\hat{\phi}_{hh}$.

Theorem 3.3.1 (Asymptotic normality of the Y-W estimator)

Let $\hat{\boldsymbol{\phi}}_{YW} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)^T$ be the Y-W estimator of $\boldsymbol{\phi}$ in a causal AR(p) model. Then

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2),$$

where Γ is the $p \times p$ matrix with $\Gamma_{ij} = \gamma(i-j)$.

The proof uses the following lemma, which is also the key to Section 3.4.

Lemma 3.3.A (Basic limits for causal AR(p); supplement)

Let $\{X_t\}$ be a causal AR(p) with $\{\epsilon_t\} \sim IID(0, \sigma^2)$, and $\mathbf{X}_{t-1} = (X_{t-1}, \ldots, X_{t-p})^T$. Then
(a) $\frac{1}{n}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s., and $\hat{\gamma}(h) \to \gamma(h)$ a.s. for each h;
(b) $\frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2\Gamma)$;
(c) Γ is nonsingular.

Proof. [Supplement]

Step 1. (a) $X_t = \sum_j a_j\epsilon_{t-j}$ is a fixed measurable function of the i.i.d. sequence shifted to time t, so $\{X_t\}$ (and any finite-lag function such as $X_{t-i}X_{t-j}$) is strictly stationary and ergodic. Birkhoff's ergodic theorem gives $n^{-1}\sum X_{t-i}X_{t-j} \to EX_{t-i}X_{t-j} = \gamma(i-j)$ and $\bar{X} \to 0$ a.s.; deleting finitely many terms does not affect the limits. The same applies to $\hat{\gamma}(h)$.

Step 2. (b) Let $\mathcal{F}_t = \sigma(\epsilon_s : s \le t)$ and fix $c \in \mathbb{R}^p$. Then $\xi_t = c^T\mathbf{X}_{t-1}\epsilon_t$ is stationary, ergodic, $\mathcal{F}_t$-measurable, and $E(\xi_t \mid \mathcal{F}_{t-1}) = c^T\mathbf{X}_{t-1}E\epsilon_t = 0$, since $\mathbf{X}_{t-1} \in \mathcal{F}_{t-1}$ and ϵ_t is independent of $\mathcal{F}_{t-1}$. Also $E\xi_t^2 = E(c^T\mathbf{X}_{t-1})^2E\epsilon_t^2 = \sigma^2c^T\Gamma c < \infty$. By the CLT for stationary ergodic martingale differences (Billingsley–Ibragimov), $n^{-1/2}\sum\xi_t \xrightarrow{d} N(0, \sigma^2c^T\Gamma c)$. The Cramér–Wold device gives (b).

Step 3. (c) This is Step 1 of the proof of Proposition 2.5.B: $c^T\Gamma c \ge c_i^2\sigma^2 > 0$ for $c \ne 0$. ∎

Proof of Theorem 3.3.1. [Supplement] Let $\hat{\Gamma} = [\hat{\gamma}(i-j)]_{i,j=1}^p$ and $\hat{\boldsymbol{\gamma}}_p = (\hat{\gamma}(1), \ldots, \hat{\gamma}(p))^T$, so $\hat{\Gamma}\hat{\boldsymbol{\phi}}_{YW} = \hat{\boldsymbol{\gamma}}_p$ (multiply (3.2) with hats by $\hat{\gamma}(0)$).

Step 1. Then $\hat{\Gamma}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) = \hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}$, whose j-th entry is $\hat{\gamma}(j) - \sum_k \phi_k\hat{\gamma}(j-k)$.

Step 2. Replace $\hat{\gamma}(l)$ by $n^{-1}\sum_t X_tX_{t+l}$ (sums over the available indices). The error is a sum of $O(p)$ boundary terms of size $O_p(1/n)$ plus terms of the form $\bar{X} \cdot n^{-1}\sum X_t$, i.e. $O_p(1/n)$ by Remark 3.1.2; after multiplying by $\sqrt{n}$ these vanish in probability. Hence

$$\sqrt{n}\Big(\hat{\gamma}(j) - \sum_k \phi_k\hat{\gamma}(j-k)\Big) = \frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\Big(X_t - \sum_k \phi_k X_{t-k}\Big) + o_p(1) = \frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\epsilon_t + o_p(1).$$

Step 3. By the Lemma, $\sqrt{n}(\hat{\gamma}_p - \hat{\Gamma}\boldsymbol{\phi}) \xrightarrow{d} N(0, \sigma^2\Gamma)$ and $\hat{\Gamma} \to \Gamma$ (nonsingular). By Slutsky,

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) = \hat{\Gamma}^{-1}\sqrt{n}(\hat{\gamma}_p - \hat{\Gamma}\boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2\Gamma\Gamma^{-1}) = N(0, \Gamma^{-1}\sigma^2).$$

■

Example 3.3.3 (Y-W estimators for AR(1) and AR(2))

For AR(1), $\rho(1) = \phi$ gives $\hat{\phi} = \hat{\rho}(1)$. For AR(2) the Y-W equations are

$$\rho(1) = \phi_1 + \rho(1)\phi_2, \qquad \rho(2) = \rho(1)\phi_1 + \phi_2,$$

so

$$\hat{\phi}_1 = \frac{\hat{\rho}(1)(1 - \hat{\rho}(2))}{1 - \hat{\rho}^2(1)}, \qquad \hat{\phi}_2 = \frac{\hat{\rho}(2) - \hat{\rho}^2(1)}{1 - \hat{\rho}^2(1)}.$$

Corollary 3.3.B (Sample PACF of an AR(p); supplement)

For a causal AR(p) and each $h > p$, $\sqrt{n}\,\hat{\phi}_{hh} \xrightarrow{d} N(0, 1)$.

Proof. [Supplement]

Step 1. An AR(p) is also an AR(h) with $\boldsymbol{\phi}^{(h)} = (\phi_1, \ldots, \phi_p, 0, \ldots, 0)^T$, and $\hat{\phi}_{hh}$ is the last component of the order-h Y-W estimator. By Theorem 3.3.1, $\sqrt{n}\,\hat{\phi}_{hh} \xrightarrow{d} N(0, \sigma^2(\Gamma_h^{-1})_{hh})$.

Step 2. For a positive definite covariance matrix, $1/(\Gamma_h^{-1})_{hh}$ is the residual variance of the last coordinate regressed linearly on the others: here $\mathrm{Var}(X_{t-h} - P(X_{t-h} \mid X_{t-h+1}, \ldots, X_{t-1}))$. By time reversal (the covariance matrix is unchanged when the order is reversed) this equals the one-step prediction error of X_t from $X_{t-1}, \ldots, X_{t-h+1}$, which is $\mathrm{Var}(\epsilon_t) = \sigma^2$ because $h - 1 \geq p$ (Proposition 2.5.B, Step 2). Hence $\sigma^2(\Gamma_h^{-1})_{hh} = 1$. ■

Thus $\pm 1.96/\sqrt{n}$ bands on the SPACF beyond lag p serve to choose the AR order.

3.4 Least squares estimators

Motivation. Least squares, the standard tool of regression, applies directly to AR models by regressing X_t on its lags. Despite the dependence of the regressors, it has the same limit distribution as in Theorem 3.3.1.

For the stationary AR(1) model $X_t = \phi X_{t-1} + \epsilon_t$, the *least squares estimator* minimizes $S(\phi) = \sum_{t=2}^{n}(X_t - \phi X_{t-1})^2$:

$$\hat{\phi} = \frac{\sum_{t=2}^{n} X_t X_{t-1}}{\sum_{t=2}^{n} X_{t-1}^2}, \qquad \sqrt{n}(\hat{\phi} - \phi) = \frac{n^{-1/2}\sum_{t=2}^{n} X_{t-1}\epsilon_t}{n^{-1}\sum_{t=2}^{n} X_{t-1}^2}.$$

In the numerator $EX_{t-1}\epsilon_t = 0$ and $\mathrm{Var}(\sum_{t=2}^{n} X_{t-1}\epsilon_t) = (n-1)\sigma^2 EX_1^2$ (the cross terms vanish because ϵ_t is independent of $X_{s-1}\epsilon_s X_{t-1}$ for $s < t$). In fact

$$\frac{1}{\sqrt{n}}\sum_{t=2}^{n} X_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2 EX_1^2) \tag{3.3}$$

(Lemma 3.3.A(b)), while the denominator tends to $\gamma(0) = EX_1^2$.

Theorem 3.4.1 (LS estimator in AR(1))

If $\hat{\phi}$ is the least squares estimator of ϕ in a stationary AR(1) model, then

$$\sqrt{n}(\hat{\phi}-\phi) \xrightarrow{d} N(0,(EX_1^2)^{-1}\sigma^2) = N(0,1-\phi^2).$$

Proof. [Supplement] By (3.3) the numerator converges in distribution to $N(0,\sigma^2 EX_1^2)$, and by Lemma 3.3.A(a) the denominator converges a.s. to $EX_1^2 > 0$. Slutsky's theorem gives the limit $N(0,\sigma^2 EX_1^2/(EX_1^2)^2)$, and $EX_1^2 = \sigma^2/(1-\phi^2)$ by Example 2.2.3. (This is also the case $p=1$ of Theorem 3.4.2.) ■

For the stationary AR(p) model $X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t$, write $\mathbf{X}_{t-1} = (X_{t-1},\ldots,X_{t-p})^T$. The least squares estimator is

$$\hat{\boldsymbol{\phi}} = (\hat{\phi}_1,\ldots,\hat{\phi}_p)^T = \Big(\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T\Big)^{-1} \sum_{t=p+1}^{n} \mathbf{X}_{t-1}X_t,$$

so that, using $X_t = \mathbf{X}_{t-1}^T\boldsymbol{\phi} + \epsilon_t$,

$$\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} = \Big(\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T\Big)^{-1} \sum_{t=p+1}^{n} \mathbf{X}_{t-1}\epsilon_t.$$

Here $E\mathbf{X}_{t-1}\epsilon_t = 0$ and $\mathrm{Cov}(\sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t) = (n-p)\Gamma\sigma^2$ with $\Gamma_{ij} = \gamma(i-j)$.

Theorem 3.4.2 (LS estimator in AR(p))

If $\hat{\boldsymbol{\phi}}$ is the least squares estimator of $\boldsymbol{\phi}$ in a stationary (causal) AR(p) model, then

$$\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0,\Gamma^{-1}\sigma^2).$$

Proof. [Supplement]

Step 1. By Lemma 3.3.A(b), $n^{-1/2}\sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0,\Gamma\sigma^2)$.

Step 2. By Lemma 3.3.A(a),(c), $n^{-1}\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s. with Γ nonsingular, so the inverse exists for large n and converges to Γ^{-1}.

Step 3. Slutsky: $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} \Gamma^{-1}N(0,\Gamma\sigma^2) = N(0,\Gamma^{-1}\sigma^2)$. ■

By Theorems 3.3.1 and 3.4.2, the Y-W and LS estimators have the same asymptotic distribution. Moreover, with $\hat{\Gamma}_{ij} = \hat{\gamma}(i-j)$ and a consistent $\hat{\sigma}^2$,

$$\frac{n(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{\hat{\sigma}^2} \xrightarrow{d} \chi^2(p), \tag{3.4}$$

because $Z_n = \sqrt{n}\,\Gamma^{1/2}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})/\sigma \xrightarrow{d} N(0,I_p)$, the statistic equals $Z_n^T Z_n + o_p(1)$ (Slutsky, as $\hat{\Gamma}\to\Gamma$, $\hat{\sigma}^2 \to \sigma^2$), and $|Z|^2 \sim \chi^2(p)$ for $Z \sim N(0,I_p)$ (continuous mapping). Hence a $100(1-\alpha)\%$ confidence region for $\boldsymbol{\phi}$ is

$$S_\alpha = \Big\{\boldsymbol{\phi}\in\mathbb{R}^p : \frac{n(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{\hat{\sigma}^2} \le \chi^2(p,1-\alpha)\Big\},$$

where $P(\chi^2(p) \ge \chi^2(p,1-\alpha)) = \alpha$.

3.5 Maximum likelihood estimators

Motivation. Maximum likelihood uses the full Gaussian joint density. For AR models the likelihood factorizes into one-step conditional densities; dropping the first factor gives the conditional MLE, which coincides with least squares.

To write a likelihood we assume $\{X_t\}$ is Gaussian, i.e. $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$; this is also the usual working assumption when the data are not normal. The likelihood equations are rarely solvable by hand, but the large-sample behaviour of the estimators is well known.

3.5.1 Exact likelihood of AR(1)

Let $X_t = \phi X_{t-1} + \epsilon_t$, $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$, $|\phi| < 1$. Then $EX_1 = 0$, $\mathrm{Var}(X_1) = \sigma^2/(1-\phi^2)$, so

$$f_{X_1}(x_1; \phi, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2/(1-\phi^2)}} \exp\Big\{\frac{-x_1^2}{2\sigma^2/(1-\phi^2)}\Big\}.$$

Given $X_1 = x_1$, since ϵ_2 is independent of X_1,

$$P(X_2 \le x \mid X_1 = x_1) = P(\phi X_1 + \epsilon_2 \le x \mid X_1 = x_1) = P(\epsilon_2 \le x - \phi x_1),$$

so $X_2 \mid X_1 = x_1 \sim N(\phi x_1, \sigma^2)$. Likewise, by the Markov structure, $f_{X_t|X_{t-1},\dots,X_1}(x_t \mid x_{t-1}, \dots, x_1) = f_{X_t|X_{t-1}}(x_t \mid x_{t-1})$ is the $N(\phi x_{t-1}, \sigma^2)$ density. Hence

$$f_{X_1,\dots,X_n}(x_1, \dots, x_n; \phi, \sigma^2) = f_{X_1}(x_1; \phi, \sigma^2) \prod_{t=2}^{n} f_{X_t|X_{t-1}}(x_t \mid x_{t-1}; \phi, \sigma^2), \tag{3.5}$$

and the log-likelihood is

$$\begin{aligned} L(\phi, \sigma^2) = &-\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\frac{\sigma^2}{1-\phi^2} - \frac{x_1^2}{2\sigma^2/(1-\phi^2)} \\ &- \frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2. \end{aligned} \tag{3.6}$$

The MLE maximizes (3.6); the score equations are nonlinear in ϕ, so numerical optimization is used. The same factorization works for AR(p) (the first factor is now the $N(0, \sigma^2 V_p)$ density of $(X_1, \dots, X_p)$, where $\sigma^2 V_p = \Gamma_p$); for ARMA(p, q) the MLE is likewise computed numerically.

3.5.2 Conditional likelihood

The *conditional maximum likelihood estimator* maximizes the density given the first observation(s), i.e. (3.5) without the factor f_{X_1}:

$$L_c(\phi, \sigma^2) = -\frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2.$$

Its maximizers are

$$\hat\phi = \frac{\sum_{t=2}^{n} X_t X_{t-1}}{\sum_{t=2}^{n} X_{t-1}^2}, \qquad \hat\sigma^2 = \frac{1}{n-1}\sum_{t=2}^{n}(X_t - \hat\phi X_{t-1})^2,$$

which are easy to compute.

Proposition 3.5.A (Conditional MLE = LS for AR(p); supplement)

For a Gaussian causal AR(p), the conditional MLE of $\boldsymbol{\phi}$ given $X_1, \dots, X_p$ equals the LS estimator $\hat{\boldsymbol{\phi}}$, the conditional MLE of σ^2 is $\hat{\sigma}^2 = (n-p)^{-1} \sum_{t=p+1}^{n} (X_t - \mathbf{X}_{t-1}^T \hat{\boldsymbol{\phi}})^2 \to \sigma^2$ a.s., and $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2)$. The exact MLE has the same limit distribution.

Proof. [Supplement]

Step 1. The conditional log-likelihood is $L_c = -\frac{n-p}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} S(\boldsymbol{\phi})$ with $S(\boldsymbol{\phi}) = \sum_{t=p+1}^{n} (X_t - \mathbf{X}_{t-1}^T \boldsymbol{\phi})^2$. For every σ^2 it is maximized in $\boldsymbol{\phi}$ by minimizing S, i.e. by the LS estimator. Then $\partial L_c / \partial \sigma^2 = -\frac{n-p}{2\sigma^2} + \frac{S(\hat{\boldsymbol{\phi}})}{2\sigma^4} = 0$ gives $\hat{\sigma}^2 = S(\hat{\boldsymbol{\phi}})/(n-p)$.

Step 2. Consistency: $S(\hat{\boldsymbol{\phi}})/(n-p) = n^{-1}\sum \epsilon_t^2 - (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T (n^{-1} \sum \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T)(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) + o(1)$ (expand $X_t - \mathbf{X}_{t-1}^T \hat{\boldsymbol{\phi}} = \epsilon_t - \mathbf{X}_{t-1}^T (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})$ and use the normal equations). The first term $\to \sigma^2$ by the strong law, the second $\to 0$ by Lemma 3.3.A and Theorem 3.4.2. The limit law of $\hat{\boldsymbol{\phi}}$ is Theorem 3.4.2.

Step 3. (Exact MLE, sketch at this level.) The exact log-likelihood is L_c plus the log-density of $(X_1, \dots, X_p)$, which does not grow with n; its derivative is $O_p(1)$, while the Hessian of L_c is of order n ($-n\Gamma/\sigma^2$ asymptotically). A one-step expansion of the score equation then shows that the exact and conditional maximizers differ by $O_p(1/n)$, so $\sqrt{n}$ times their difference tends to 0 and the limits coincide. ■

3.6 Prediction

Motivation. Forecasting is the ultimate goal of time series analysis. The mean-square optimal predictor is a conditional expectation; for ARMA models it can be computed recursively from the model equation.

Given $X_1, \dots, X_n$, the *predictor* of X_{n+h} is

$$f(X_1, \dots, X_n) := E(X_{n+h} \mid X_1, \dots, X_n).$$

Proposition 3.6.A (Optimality of the conditional mean; supplement)

If $EX_{n+h}^2 < \infty$, then $E\big(X_{n+h} - g(X_1, \dots, X_n)\big)^2$ over all measurable g with $Eg^2 < \infty$ is minimized by $g = f$.

Proof. [Supplement] Write $Y = X_{n+h}$, $\mathbf{X} = (X_1, \dots, X_n)$. Then

$$E(Y - g(\mathbf{X}))^2 = E(Y - f(\mathbf{X}))^2 + E(f(\mathbf{X}) - g(\mathbf{X}))^2 + 2E\big[(Y - f(\mathbf{X}))(f(\mathbf{X}) - g(\mathbf{X}))\big].$$

Conditioning on $\mathbf{X}$, the cross term is $2E\big[(f(\mathbf{X}) - g(\mathbf{X}))\, E(Y - f(\mathbf{X}) \mid \mathbf{X})\big] = 0$. Hence $E(Y-g)^2 \geq E(Y-f)^2$, with equality iff $g(\mathbf{X}) = f(\mathbf{X})$ a.s. ■

If $\{X_t\}$ is Gaussian, f is a linear combination $\sum_{i=1}^{n} a_i X_i$ (Section 1.2), and the a_i minimize $E(X_{n+h} - \sum_i a_i X_i)^2$. In general f is not linear, so the best linear predictor is not optimal in the strict sense; but since f is hard to find, linear predictors are used in practice. For stationary series efficient algorithms for the coefficients exist (e.g. the Durbin–Levinson algorithm), and for ARMA models the predictor is a function of the model parameters.

For forecasting from the infinite past write $\hat{X}_t(h) = E(X_{t+h} \mid X_t, X_{t-1}, \dots)$.

Example 3.6.1 (Forecasting an AR(1) with mean)

Let $(1-\phi B)(X_t-\mu)=\epsilon_t$, $\{\epsilon_t\}\sim IID(0,\sigma^2)$, $|\phi|<1$, so $X_t=\mu+\phi(X_{t-1}-\mu)+\epsilon_t$. Since ϵ_{t+1} is independent of $X_t, X_{t-1},\dots$,

$$\hat{X}_t(1)=E(\mu+\phi(X_t-\mu)+\epsilon_{t+1}\mid X_t,X_{t-1},\dots)=\mu+\phi(X_t-\mu).$$

Similarly $\hat{X}_t(h)=\mu+\phi(\hat{X}_t(h-1)-\mu)$, hence

$$\hat{X}_t(h)=\mu+\phi^h(X_t-\mu)\longrightarrow\mu\quad(h\to\infty).$$

In practice μ,ϕ are replaced by $\hat{\mu},\hat{\phi}$.

For a stationary invertible ARMA(1,1), $(1-\phi B)(X_t-\mu)=(1+\theta B)\epsilon_t$ with $|\phi|<1$, $|\theta|<1$, we have $X_{t+1}=\mu(1-\phi)+\phi X_t+\epsilon_{t+1}+\theta\epsilon_t$. By invertibility ϵ_t is a function of $X_t, X_{t-1},\dots$, so $E(\epsilon_t\mid X_t,X_{t-1},\dots)=\epsilon_t$ and

$$\hat{X}_t(1)=\mu(1-\phi)+\phi X_t+\theta\epsilon_t,\qquad \hat{X}_t(h)=\mu(1-\phi)+\phi\hat{X}_t(h-1),\quad h\geq 2.$$

In practice ϵ_t is unknown and is replaced by the residual $\hat{\epsilon}_t$, and μ,ϕ,θ by their estimates; the forecasts are then computed recursively from $\hat{X}_t(1)$.